

Introduction to Impact Evaluation: Key Concepts and Terminology

Module 2 · Day 1, Session 1: Building a Common Language for Causal Analysis

1 · What are we comparing?

The Al Falah Road Safety Campaign

A national government runs a one-year awareness campaign: billboards, TV spots, and school talks promoting seatbelt use and safe speeds.

At the end of the year, someone has to decide: did it work, and should it run again?

This is an illustrative example, not a real Abu Dhabi case study.

Ran	One year, nationwide
Delivered	Billboards, TV spots, school talks
Outcome	Traffic fatalities
The question	Did the campaign save lives?

The headline number

-30%

TRAFFIC FATALITIES, YEAR ON YEAR

DEATHS, YEAR BEFORE **420**

DEATHS, YEAR AFTER **294**

FEWER DEATHS **126**

Many times we make decisions based on numbers

Your first decision. The programme reported that traffic fatalities fell by 30% in one year.

Would you fund a second year?

Go to **menti.com** and enter code **1234 5678**, or scan the code on the right. Please, give an answer.

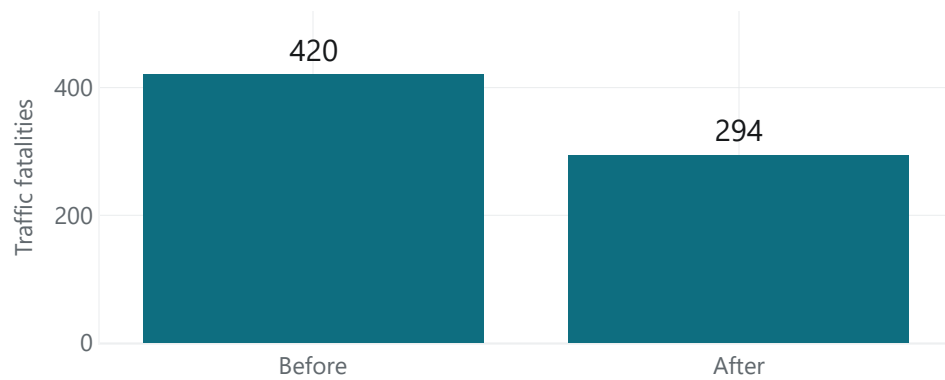


Every number you will be shown this week is **correct**. Almost none of them will be *wrong*.

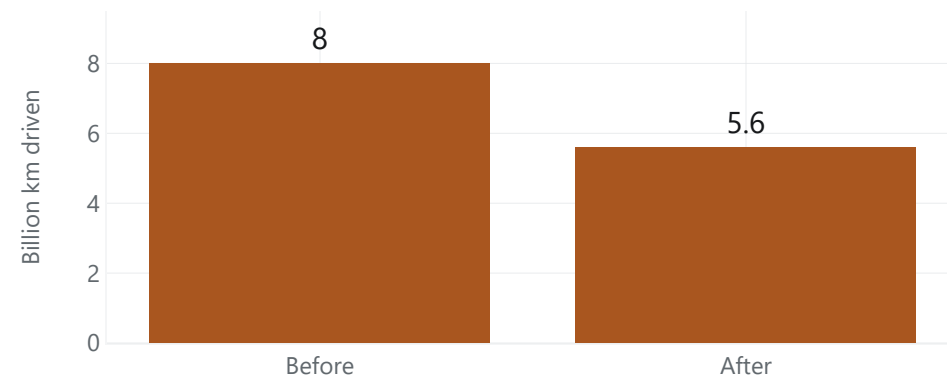
BUT... sometimes an accurate number answers a question you did not ask, so we might still be wrong.

What else happened that year

Fatalities, nationwide



Distance driven, nationwide



Fuel prices rose sharply that year, and drivers cut back: distance driven fell by 30%, almost exactly as much as fatalities did.

World A · the same headline, a real effect

-30%

DEATHS PER BILLION KM DRIVEN

RATE BEFORE **52** RATE AFTER **37** THE CAMPAIGN WORKED **-30%**

World A is not what happened. It is what the **same headline** would have meant if driving had not changed at all: every one of those 126 fewer deaths would be the campaign's doing, and the death rate would have fallen by 30%.

World B · the death rate did not move

0%

DEATHS PER BILLION KM DRIVEN

RATE BEFORE **52** RATE AFTER **52** LIVES SAVED BY THE CAMPAIGN **0**

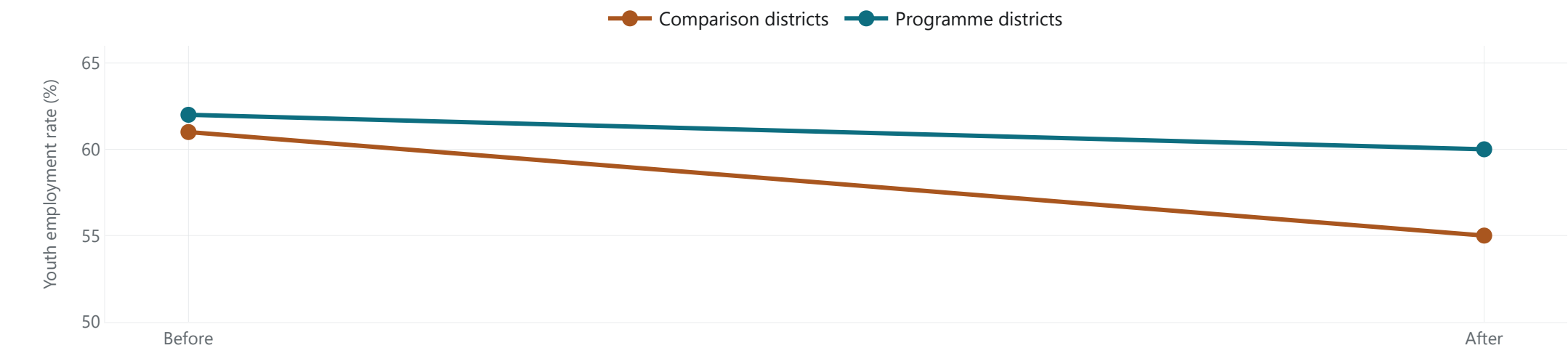
If the roads emptied by 30% and deaths fell by 30%, the risk of dying on them did not change. The campaign saved nobody. The headline was the fuel price.

Two worlds, one headline, opposite verdicts. Nothing in the headline, or in the number 30%, tells you which world you are in.

Did it work?



The trap also runs the other way



A youth employment programme: employment fell 2 points in the districts it ran in. On a before-and-after reading, it failed. But employment fell 6 points in the comparison districts, so the programme held up 4 points that would otherwise have been lost.

"Compared to what?"

Every evaluation result is a comparison. A before-and-after number compares the programme to *the past*, not to what would have happened anyway. The comparison can move the answer down, as in World B, or up, as in the mirror case. The direction is not knowable in advance. If you do not know what a number is being compared to, you do not yet know what it means.

2 · Three questions for any number

One idea, three questions

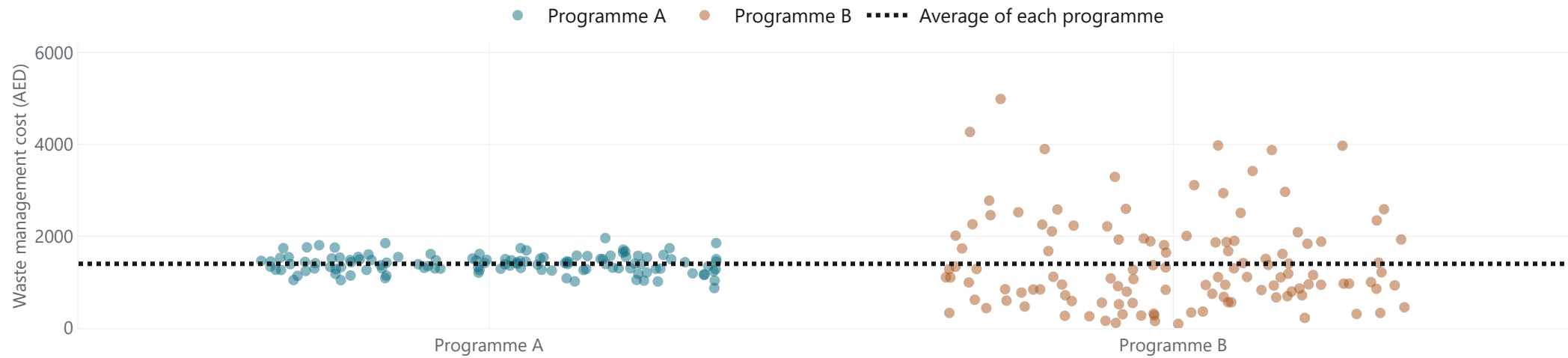
Five words follow. They are easier to hold if you attach them to the three questions you are entitled to ask of any number you are handed.

Compared to what?	Comparison group · the question nobody puts in the summary
How big, in units that matter?	Mean · treatment effect · coefficient
How sure are we?	p-value · confidence interval

Keep these three questions. You will use them on every slide for the rest of the week, and on the next evaluation report that lands on your desk.

3 · The five words

Word 1 · Mean



Each dot is one business. Both programmes average exactly 1,400 AED

In Programme A, 59% of businesses are within 10% of the average. In Programme B, only 10%.

Which would you fund, and what else do you need to know?

Word 2 · Treatment effect

Programme schools	54% → 71%
Comparison schools	50% → 65%
Decision rule	Must add 3 points to justify scale-up

Grade 6 maths pass rates.

Three numbers describe this result: 17, 6 and 2.

Work out what each one is, then decide which is the treatment effect.

	Before	After	Change
Programme schools	54	71	17
Comparison schools	50	65	15
Difference	4	6	2

The bottom row is the two above it subtracted.

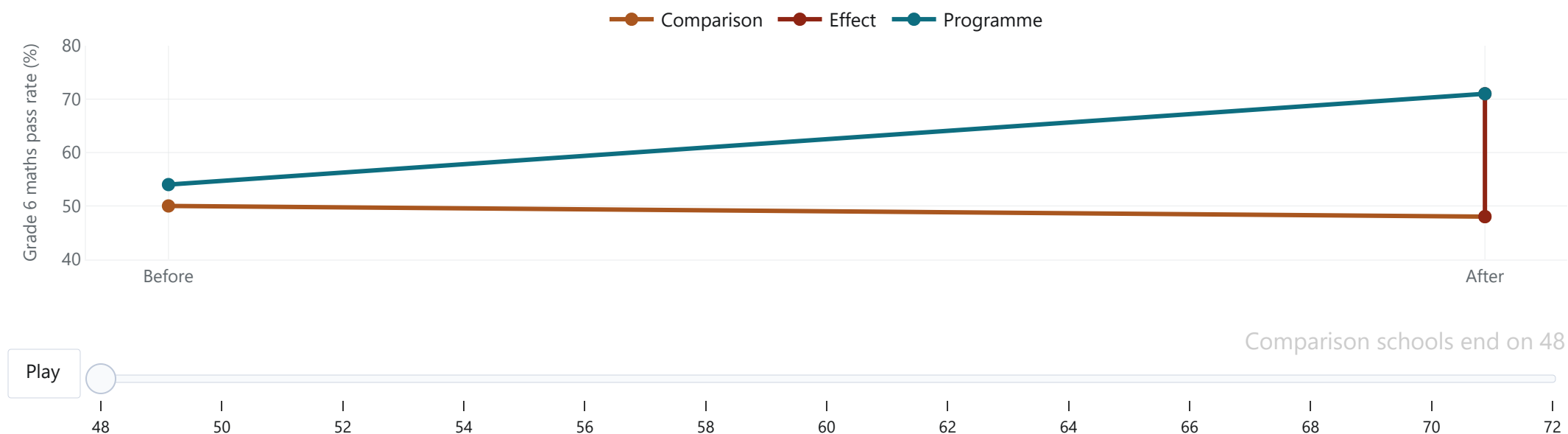
17 is not the answer. The national reform was lifting both groups.

6 is not the answer either. It carries the 4-point head start the programme schools already had.

2 is the treatment effect: 17 minus 15, and equally 6 minus 4.

Word 2 · The effect depends on the comparison

The programme schools are fixed at 54% then 71%. Only the comparison schools move. The red bar is the treatment effect, and it is the only thing on the chart that responds.



Press play, or drag the slider. Programme schools never move. Which comparison value makes the effect zero, and what does that value mean in plain words?

Word 3 · Coefficient

A coefficient reports the size of a relationship as a **rate**: how much the outcome moves per one unit of whatever sits on the row. On an evaluation table, the coefficient on the programme row is often the treatment effect itself.



Hover the diamonds, then the red bar. Here the coefficient is the vertical gap between the two group means, 3.0 points. That gap is what the model reports on the programme row of the next slide.

Reading the table column by column

Variable	Coefficient	Std. error	p-value	95% CI
(Intercept)	40.50	3.20	<0.001	[34.20, 46.80]
Programme (enrolled = 1)	2.10	0.90	0.021	[0.30, 3.90]
Baseline pass rate	0.58	0.04	<0.001	[0.50, 0.66]
Class size (students)	-0.32	0.14	0.024	[-0.59, -0.05]
Teacher experience (years)	0.15	0.08	0.06	[-0.01, 0.31]

N = 240 schools. Outcome: Grade 6 maths pass rate.

Programme row: the one number that answers the decision. 2.1 points, interval [0.3, 3.9]. The rule needs 3 points. The interval contains 3, so we cannot say whether the programme clears the bar.

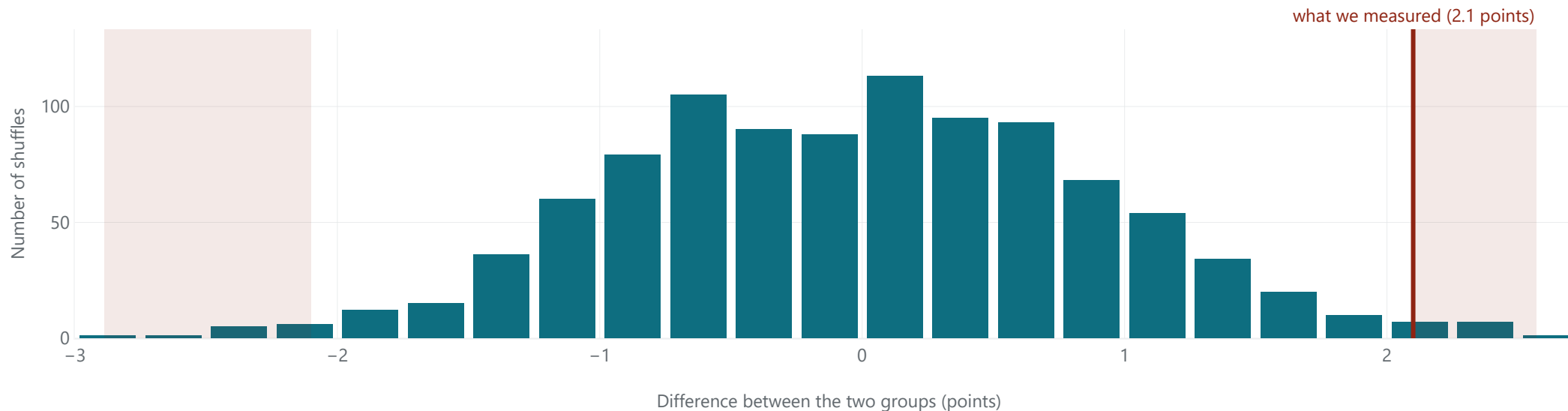
Intercept: the value when everything on the table is zero. Nobody is a school with zero students and zero teachers. Ignore it.

Covariate rows: class size, teacher experience, baseline pass rate. These adjust the comparison. They are not recommendations: the coefficient on teacher experience is 0.15, and its interval [-0.01, 0.31] crosses zero, so it cannot rule out no effect at all.

Std. error: the wobble. Divide the coefficient by it to see how steady the estimate is.

Word 4 · p-value

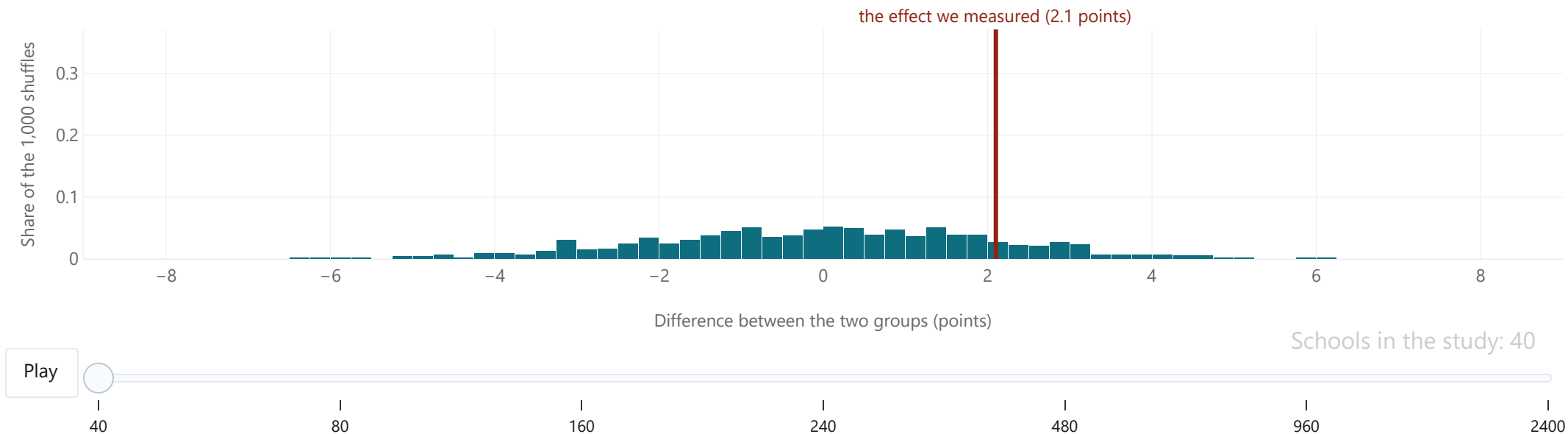
Plain language: roughly, how surprising this result would be if the programme had truly done nothing at all. A small p-value means “this would be a strange coincidence if there were no effect.”



We took the 240 schools, broke the link between the programme and the outcome, and did it 1,000 times. The red tails are the shuffles that beat what we measured: 0.023, or about 23 of 1,000. That fraction is the p-value.

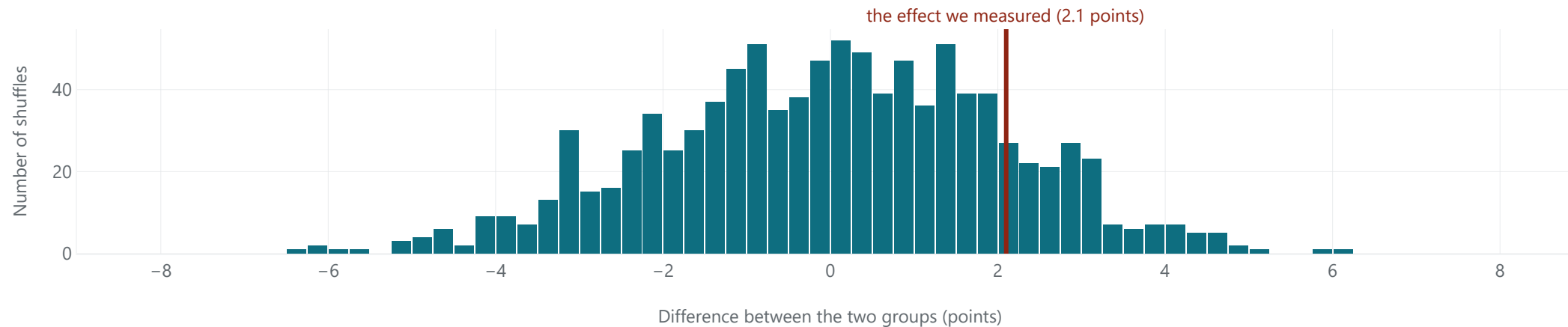
Word 4 · Why significance is not importance

Same programme. Same effect of 2.1 points. Three studies of different sizes. Press play, or drag the slider, and watch the shuffled distribution tighten around zero. The effect never changes.



At 40 schools the same effect gives $p = 0.32$. At 240 schools, $p = 0.02$. At 2,400 schools it is below 0.001. The programme did not improve by a single point. **Statistically significant does**

Word 4 · The warning cuts both ways



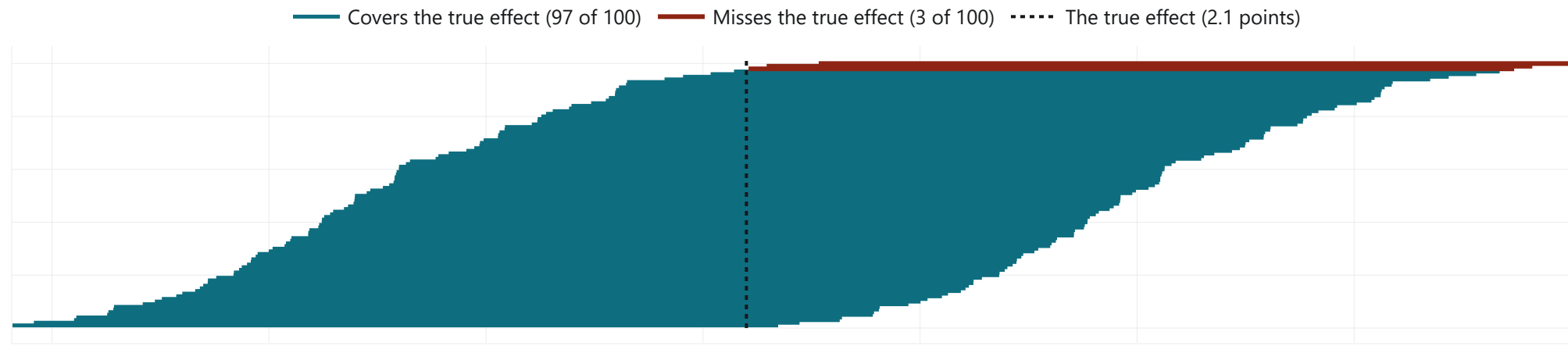
At 40 schools the same study would have found nothing, and nothing is exactly what it would have reported. **Absence of evidence is not evidence of absence.**

An evaluator reports “no significant effect”. Before you accept it, ask two things. How many businesses were in the study? And what is the smallest effect the study could have detected?

Word 5 · Confidence interval

Plain language: a range of plausible values for the true effect, not one precise number. A wider range means less certainty, not a bigger effect.

10 repeat studies of the same programme



Hover any bar for one study's estimate. Click a legend entry to isolate the misses. Now the decision. All three programmes report an effect of 2.1 points, and the bar to clear is 3 points.

A [0.3, 3.9] **B** [1.9, 2.3] **C** [-0.6, 4.8]

Which one would you act on, and what is your answer for each?

B is precise and below the bar: confidently not enough. A is too uncertain to call. C is worse still. **The interval decides, not the point estimate.** An interval that spans the decision threshold means the honest answer is that we do not yet know.

4 · Put AI to work

AI prompt cards

Each group takes one card. Paste its prompt into your AI tool, read the answer together, and mark anything you think it gets wrong or overstates. Then compare with the answer on the next slide.

Prompt: "Explain what a 95% confidence interval means, in one paragraph, for someone who is not a statistician."

AI: "A 95% confidence interval is the range in which the true value is likely to fall. **There is a 95% probability that the true effect lies inside this interval**, so if the interval is [0.3, 3.9] the true effect is probably between 0.3 and 3.9. A narrower interval means the estimate is more precise."

Card 2 "Explain what a p-value means."

Card 3 "School pass rates rose 5 percentage points after our programme. Is that a good result?"

Debrief

Card 1. The flagged sentence is the classic error. A 95% interval does not carry a 95% probability for *this* interval. It means that if you ran the study 100 times, 95 of the intervals would catch the true value, as the previous chart shows. The AI stated a probability the method does not deliver.

Card 3. The AI almost certainly answered without asking what the pass rate was compared to, and without asking what bar the programme has to clear. Both are missing from the prompt, and a fluent answer hides that.

Card 4. Look for a recommendation with no interval. If the AI tells you to reduce class sizes without noting that the interval decides whether the relationship is real, it has answered a different question.

5 · Terms bingo

Terms bingo

1. "Waste costs in participating businesses fell 30% over the programme period."

2. "The programme effect is 1,240 AED, with a 95% confidence interval of 210 to 2,270."

3. "The improvement is statistically significant ($p = 0.008$), so the programme should be scaled nationally."

4. "Costs fell 420 AED in the programme group while rising 90 AED in the comparison group."

5. "Across the sample, the average business saved 1,449 AED."

6. "With 40 businesses in the study, no effect was detected."

7. "The coefficient on class size is -0.32, so reducing class size raises pass rates."

8. "Costs fell 420 AED against a rise of 90 AED in the comparison group. The programme must cut costs by at least 300 AED to justify scale-up."

Mark every term you hear used. Then make two calls, using the three questions.

Excerpt 3: would you act on it?

Excerpt 8: would you act on it? They do not have the same answer.

6 · Take this back to your desk

The three questions

From memory, before you look: what are the three questions you can ask of any number?

- 1 Compared to what?
- 2 How big is it, in units that matter to the decision?
- 3 How sure are we?